

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Database Management Systems
COURSE CODE: CSA05

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Technical Presentation (Low)
Assessment Tool Weightage: 10%

QUESTIONS		
Q. No.	Question	Bloom's Level
1	Create and present an ER diagram for an Online Library System, explaining each component clearly.	BL2 – Understand
2	Prepare a presentation on the EER model, demonstrating generalization and specialization with a University case study.	BL2 – Understand
3	Present the concept of data independence (logical and physical) and its significance in DBMS architecture.	BL2 – Understand
4	Deliver a technical presentation on the software components of a DBMS, explaining query processor, storage manager, and transaction manager.	BL1 – Remember
5	Prepare a presentation comparing strong and weak entities in ER modeling with real-world examples.	BL2 – Understand

Code of Conduct:

*I **Sandra Sudevan(192424422)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

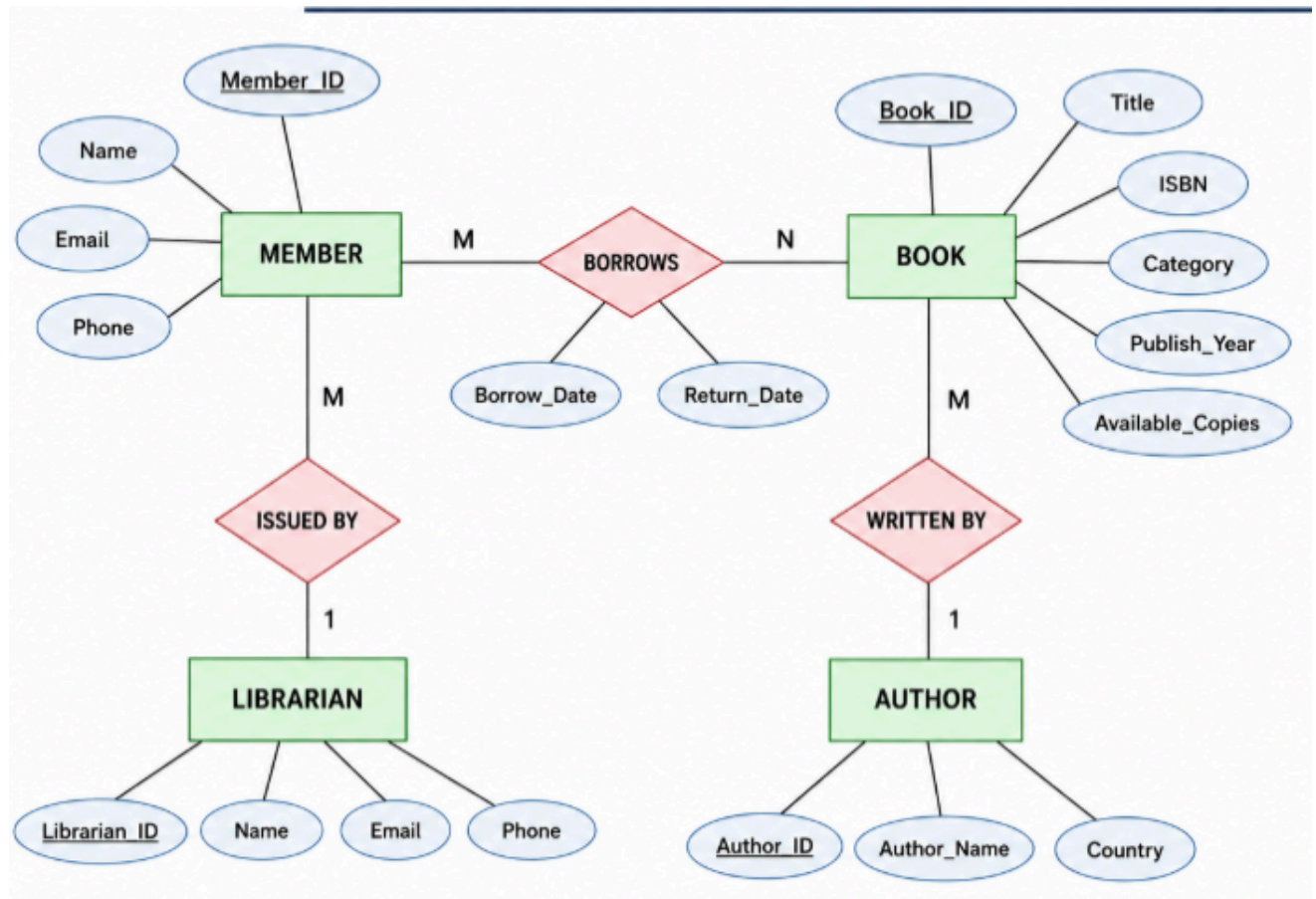
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

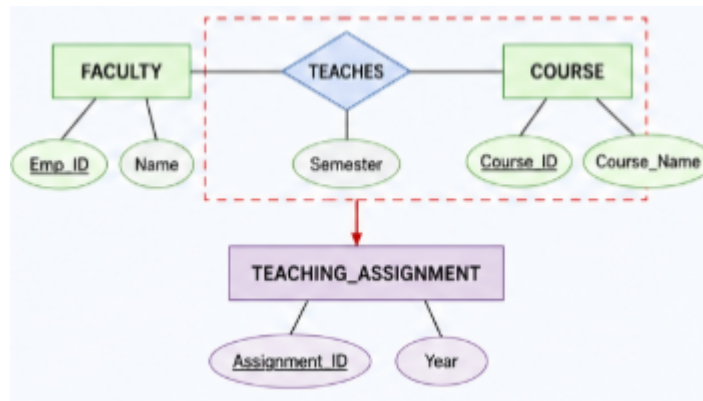
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Q1. ER Diagram for an Online Library System



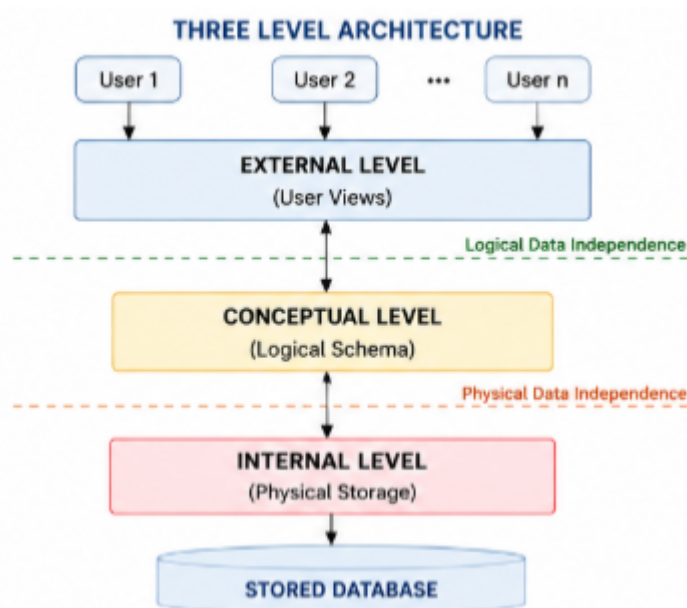
- Introduction: Online Library System manages books, members, borrowing and returns.
- Entities: Member, Book, Librarian, Author, Borrow
- Attributes: Member(Member_ID, Name, Email), Book(Book_ID, Title, ISBN), Author(Author_ID, Author_Name), Librarian(Librarian_ID, Name)
- Relationships: Member borrows Book (M:N), Author writes Book (1:M), Librarian issues Book (1:M)
- Conclusion: ER diagrams help design databases by identifying entities, attributes and relationships.

Q2. EER Model – University Case Study



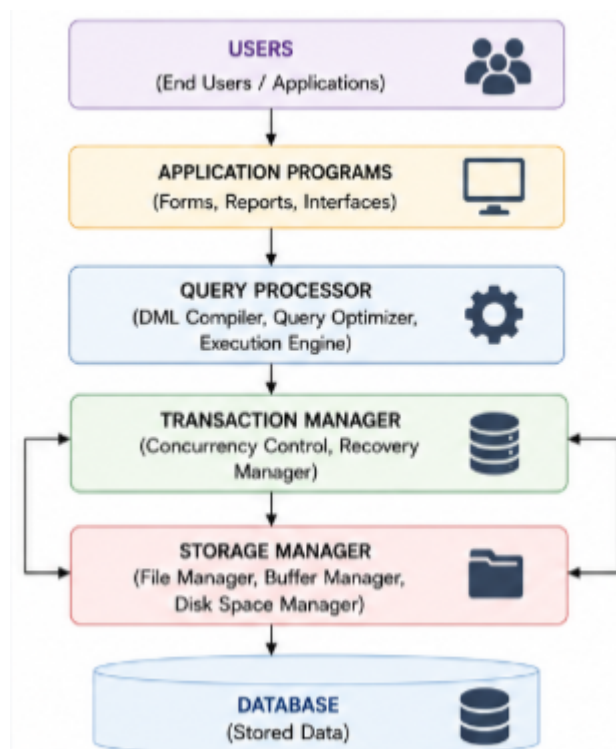
- Introduction: EER extends ER using Generalization, Specialization, Inheritance and Aggregation.
- Superclass: Person
- Subclasses: Student, Faculty, Staff
- Generalization combines common entities into Person.
- Specialization divides Person into Student, Faculty and Staff.
- Conclusion: EER models represent complex systems accurately.

Q3. Data Independence in DBMS



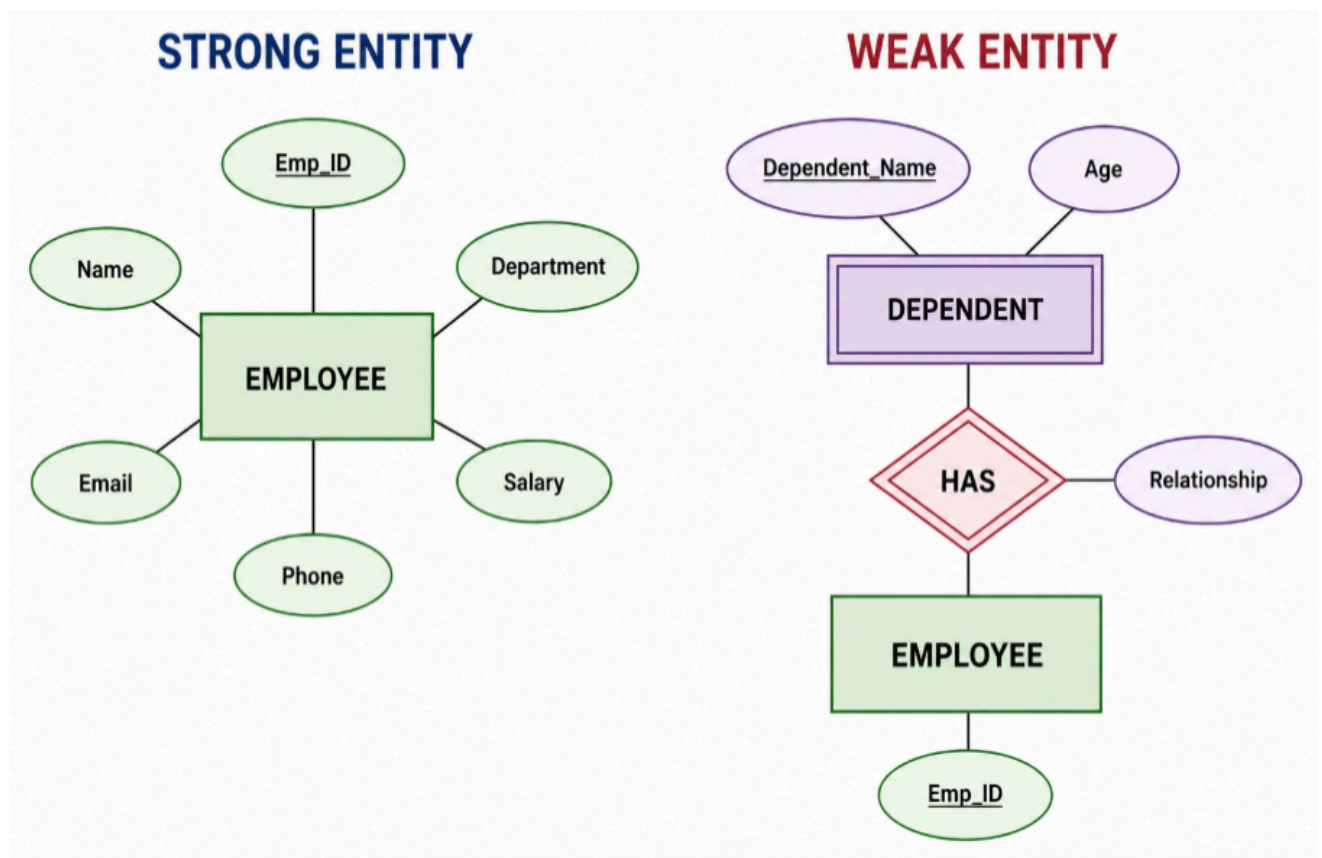
- Definition: Ability to modify one database level without affecting higher levels.
- Physical Data Independence: storage changes don't affect logical schema.
- Logical Data Independence: logical changes don't affect applications.
- Advantages: flexibility, maintenance, scalability.
- Conclusion: Improves database maintainability.

Q4. Software Components of DBMS



- Query Processor: parses, optimizes and executes SQL.
- Storage Manager: stores, retrieves and manages files.
- Transaction Manager: maintains ACID, concurrency and rollback.
- Architecture: User → Application → Query Processor → Transaction Manager → Storage Manager → Database.
- Conclusion: Components ensure secure and efficient DBMS operation.

Q5. Strong and Weak Entities



- Strong Entity: has primary key and independent (e.g., Student).
- Weak Entity: depends on strong entity (e.g., Dependent).
- Comparison: Strong has PK; Weak uses identifying relationship.
- Examples: Strong–Employee, Customer. Weak–Dependent, Order Item.
- Conclusion: Weak entities require strong entities for identification.